Why Can't We Just Trust the Table?

Example: Find the following limit, if it exists.

$$\lim_{x \to 0} \sin \frac{\pi}{x}$$

x	$\sin \frac{\pi}{x}$
0.1	
0.01	
0.001	
0.0001	
0.00001	
0.000001	

x	$\sin \frac{\pi}{x}$
0.3	
0.03	
0.003	
0.0003	
0.00003	
0.000003	

x	$\sin \frac{\pi}{x}$
0.7	
0.07	
0.007	
0.0007	
0.00007	
0.000007	

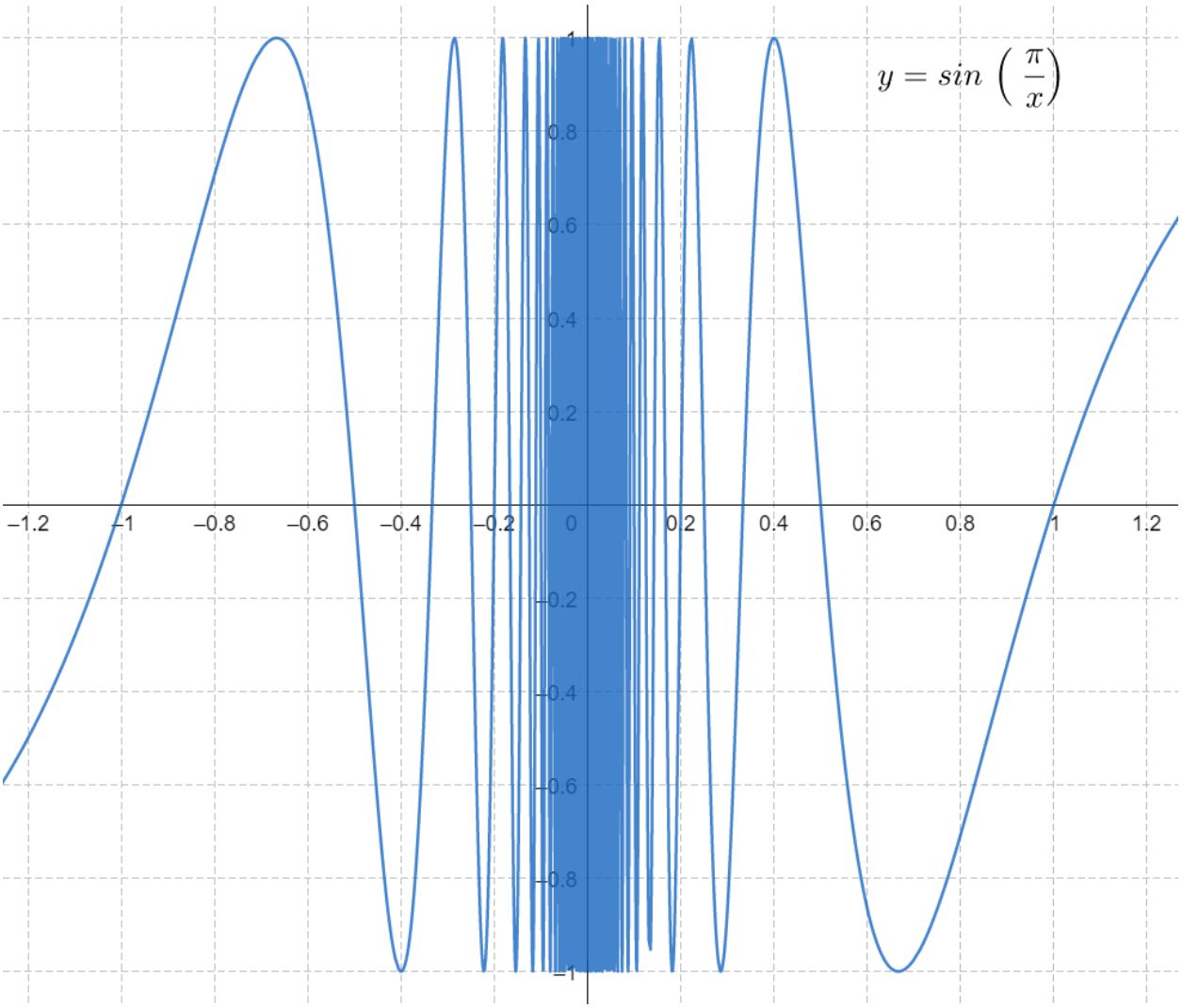
x	$\sin \frac{\pi}{x}$
0.9	
0.09	
0.009	
0.0009	
0.00009	
0.000009	

x	$\sin\frac{\pi}{x}$
0.1	0
0.01	0
0.001	0
0.0001	0
0.00001	0
0.000001	0

x	$\sin\frac{\pi}{x}$
0.3	$-\frac{\sqrt{3}}{2}$
0.03	$-\frac{\sqrt{3}}{2}$
0.003	$-\frac{\sqrt{3}}{2}$
0.0003	$-\frac{\sqrt{3}}{2}$
0.00003	$-\frac{\sqrt{3}}{2}$
0.000003	$-\frac{\sqrt{3}}{2}$

x	$\sin\frac{\pi}{x}$
0.7	– 0.974297912
0.07	0.781831482
0.007	0.433883739
0.0007	0.974927912
0.00007	– 0.781831482
0.000007	– 0.433883739

x	$\sin\frac{\pi}{x}$
0.9	– 0.342020143
0.09	– 0.342020143
0.009	– 0.342020143
0.0009	– 0.342020143
0.00009	– 0.342020143
0.000009	– 0.342020143


$y = sin\left(\frac{\pi}{x}\right)$
1
0.8
0.6
0.4
0.2
−1.2
−1
−0.8
−0.6
−0.4
−0.2
0
0.2
0.4
0.6
0.8
1
1.2
−0.2
−0.4
−0.6
−0.8
−1